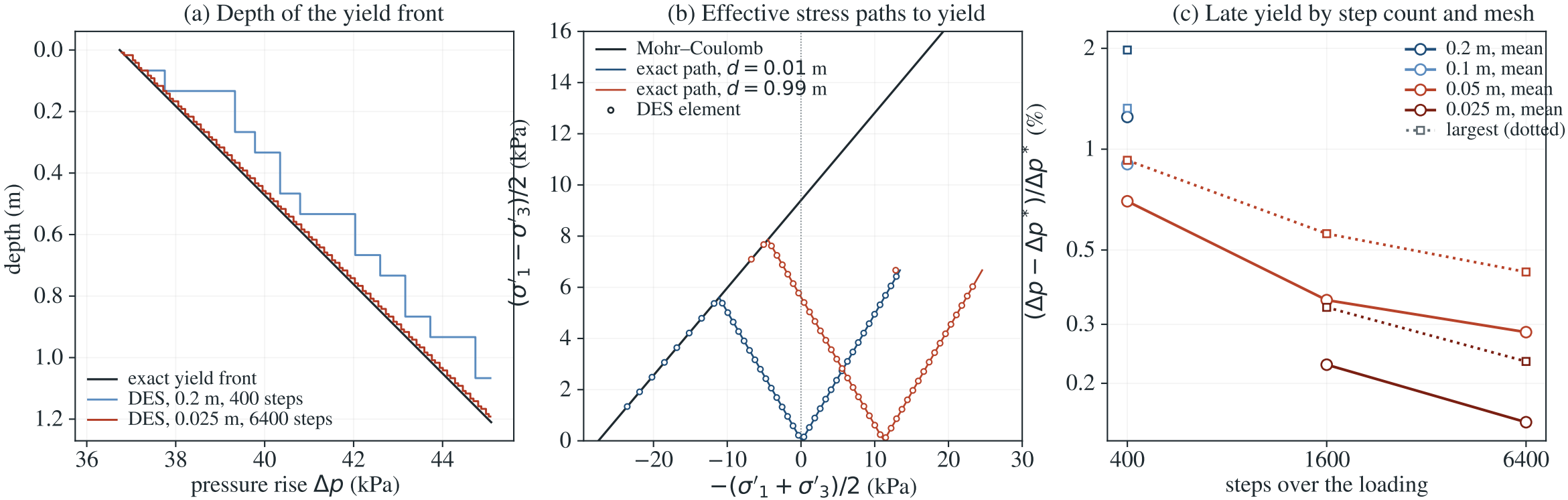

(a) Depth of the yield front
depth (m)
pressure rise Δp (kPa)
exact yield front
DES, 0.2 m, 400 steps
DES, 0.025 m, 6400 steps
(b) Effective stress paths to yield
(σ′1 − σ′3)/2 (kPa)
−(σ′1 + σ′3)/2 (kPa)
Mohr–Coulomb
exact path, d = 0.01 m
exact path, d = 0.99 m
DES element
(c) Late yield by step count and mesh
(Δp − Δp*)/Δp* (%)
steps over the loading
0.2 m, mean
0.1 m, mean
0.05 m, mean
0.025 m, mean
largest (dotted)